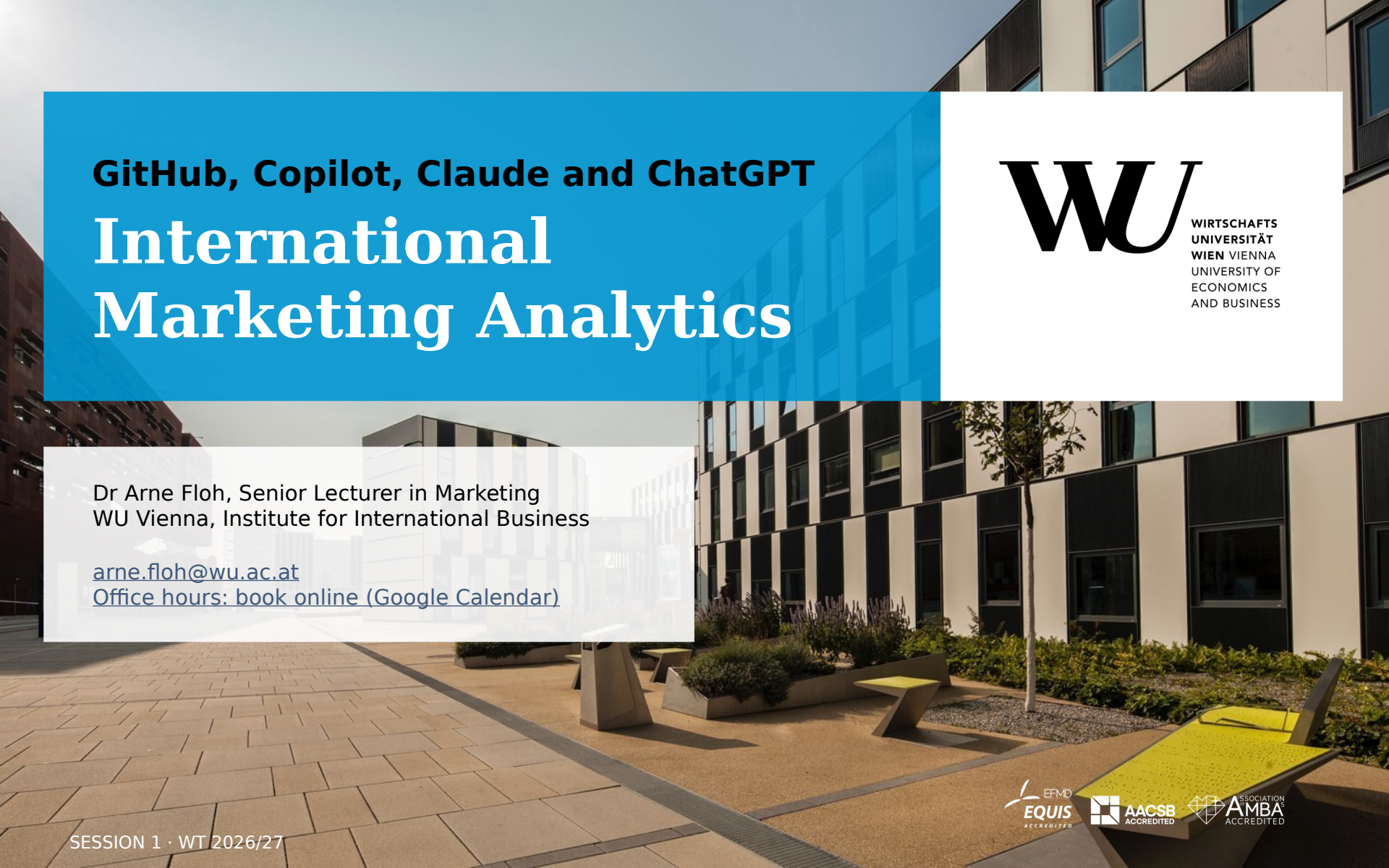
GitHub, Copilot, Claude and ChatGPT

International Marketing Analytics

WIRTSCHAFTS UNIVERSITÄT WIEN VIENNA UNIVERSITY OF ECONOMICS AND BUSINESS

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SESSION 1 · WT 2026/27

Today: GitHub and AI assistants, step by step

Account → Education → Desktop → Clone and sync → Copilot → Claude, ChatGPT

GitHub, 1–6

- Account and GitHub Education
- GitHub Desktop
- Clone, pull, commit, push
- Private or public

Copilot, 7–12

- Free student plan
- Connect it to Positron
- Chat and completions
- Use it well

Optional, 13–18

- Claude and ChatGPT
- API key or subscription
- Claude Code and Codex
- Agents in your session

Course default: GitHub Copilot, free with GitHub Education. Claude and ChatGPT are optional.

Git, GitHub and GitHub Desktop

Git

The engine: records every saved version of a folder

GitHub

The website: keeps a copy online and lets others access it

GitHub Desktop

The app: runs Git with buttons and brings its own Git

Word	Meaning	Button in GitHub Desktop
Repository (repo)	a project folder with its full history	Current repository
Clone	download a repo to your laptop, linked to GitHub	File → Clone repository
Commit	save a snapshot with a short message	Commit to main
Push / pull	send your commits up / get new commits down	Push origin / Pull origin
Fetch	check GitHub for new commits, change nothing	Fetch origin

Your GitHub account

Part 1 of 18 · sign up, two-factor sign-in, a private e-mail address

Create your GitHub account

1. Open github.com/signup
2. **E-mail:** your WU student address
3. **Password** of at least 15 characters, or 8 with a number and a lower-case letter
4. **Username:** short and professional, e.g. anna-berger; it is public
5. **Verify:** type the code GitHub sends to your inbox
6. **Plan:** choose Free; skip the questions if you like

Good to know

- One account per person: keep it for life
- Free plan: unlimited public and private repos
- Your username appears on every commit

Have an account? Keep it; add your WU address under Settings → Emails.

Secure it: two-factor authentication (2FA)

Turn on 2FA

1. **Profile picture** → Settings
2. **Password and authentication** → Enable two-factor authentication
3. **Scan the QR code** with an authenticator app
4. **Type** the 6-digit code from the app
5. **Download** your recovery codes and keep them safe
6. **Add a backup:** GitHub Mobile or a passkey

45
days to set it up once GitHub asks

TOTP
app recommended as main method

16
recovery codes: your way back in

Apps that work: Microsoft Authenticator, Google Authenticator, 1Password, or GitHub Mobile as backup.

Keep your e-mail address private

In your GitHub settings

1. **Settings** → Emails
2. **Add** your WU address if it is missing, and verify it
3. **Tick** Keep my email addresses private
4. **Tick** Block command line pushes that expose my email
5. **Copy** the noreply address shown there

Your noreply address (example)

```
1234567+anna-berger@
  users.noreply.github.com
```

Public commits show the commit e-mail to anyone

The noreply address still links commits to you

WU address: keep it verified on your account. GitHub Education checks it, even when your commits use the noreply address.

GitHub Education

Part 2 of 18 · who qualifies, what you get, how to apply

GitHub Education: what students get

GitHub Copilot

- Free for verified students, with premium features
- Set up in Parts 7 to 12

Codespaces

- Cloud computers: up to 180 core hours a month
- Not needed for this course

Developer Pack

- Free offers from partner companies
- Tools, domains, learning platforms

Portal: education.github.com shows your benefits once you are verified

Time-limited: GitHub re-checks your status; benefits end when you are no longer a student

Sources: GitHub Docs, About GitHub Education for students (2026)

Who qualifies, and which proof works

You qualify if you

- Study in a degree programme (WU: yes)
- Have your own personal GitHub account
- Are at least 13 years old
- Can prove current enrolment

Proof GitHub accepts

- Student ID showing the current enrolment date
- Enrolment confirmation (Studienbestätigung)
- Transcript of records
- Your class schedule

Academic e-mail first: add and verify your WU address in GitHub before you apply; lower case, no plus sign (anna+x@...).

Sources: GitHub Docs, Apply to GitHub Education as a student (2026)

Apply in six steps

1. Open github.com/settings/education/benefits
2. **Click** Start an application; choose Student
3. **School:** Vienna University of Economics and Business
4. **E-mail:** select your verified WU address
5. **Proof:** upload or photograph your document
6. **Submit** and wait for the e-mail with the decision

Minutes to several days: carry on with Part 3 meanwhile.

Photo tips

- Whole document in the frame
- Name, school and current date readable
- Daylight, no glare
- Allow camera and location if the browser asks
- Check the status on the same page

Not approved? Common reasons and fixes

Reason	What to do
Blurry document, or no current date	Reapply with a sharp photo of a dated enrolment confirmation
WU address not added or not verified	Settings → Emails: add it, click the link in the confirmation e-mail
Address with capitals or a plus sign	Delete it, add it again in lower case without +
Address already used by another account	Use the account that got the pack; merge accounts
Still stuck	support.github.com/contact/education, describe each step you tried

Reapply any time: same page, Start an application.

Install and set up GitHub Desktop

Part 3 of 18 · download, sign in, your name and e-mail for commits

Install GitHub Desktop

Windows 10/11 (64-bit)

1. Open desktop.github.com
2. **Click** Download for Windows
3. **Double-click** the setup file in Downloads
4. **Wait:** GitHub Desktop opens by itself

macOS 12 or later

1. Open desktop.github.com
2. **Click** Download for macOS
3. **Double-click** the zip file, then drag the app to Applications
4. **Open** GitHub Desktop from Applications

Git included: you do not need to install Git separately

Updates: GitHub Desktop updates itself; restart it when it says so

Sign in to GitHub in GitHub Desktop

Open the Accounts pane

1. **Windows:** File → Options → Accounts
2. **macOS:** GitHub Desktop → Settings → Accounts
3. **Click** Sign in to GitHub.com
4. **Click** Continue with browser

In the browser

1. **Sign in** with username and password
2. **Enter** the 2FA code from your app
3. **Click** Authorize desktop
4. **Allow** the browser to open GitHub Desktop

Done when: your name and avatar appear in the Accounts pane.

Your name and e-mail for commits

Options / Settings → Git

1. **Windows:** File → Options → Git
2. **macOS:** GitHub Desktop → Settings → Git
3. **Name:** your real name, e.g. Anna Berger
4. **Email:** pick the …@users.noreply.github.com address
5. **Default branch:** leave it at main; click Save

Name

```
Anna Berger
```

Email

```
1234567+anna-berger@users…
```

Default branch

```
main
```

Privacy: with the noreply address your WU or private e-mail never shows in a public repository.

Clone a repository



Part 4 of 18 · two ways to clone, where to put the folder, opening it in Positron

Clone: two ways, same result

A From github.com

1. **Open** the repository page on github.com
2. **Click** the green Code button
3. **Click** Open with GitHub Desktop
4. **Choose…** a local folder, then Clone

B In GitHub Desktop

1. **File →** Clone repository
2. **Tab URL:** paste the repository address
3. **Local path:** Choose… a folder
4. **Click** Clone

The course repository (public: no sign-in needed to clone)

```
https://github.com/arnefloh-wu/international-marketing-analytics-teaching
```

Where the folder should live

Good places

- Windows: C:\Users\anna\wu\ima
- macOS: /Users/anna/wu/ima
- Short path, no spaces, no umlauts
- On the laptop's own disk

Avoid

- OneDrive, iCloud Drive, Dropbox folders
- Desktop and Documents if they sync to the cloud
- Paths like …/Mein Kurs (WU)/
- USB sticks and network drives

Find the folder later: Repository → Show in Explorer (Windows) / Show in Finder (macOS)

Moved it by mistake? GitHub Desktop says it cannot find it: click Locate… and pick the new place

Open the clone in Positron

Clone in GitHub Desktop → Positron: Open Folder → Pick the cloned folder → Work, save → Back to GitHub Desktop

Positron

Edit and run code, render Quarto files.

Saved changes show up in GitHub Desktop at once.

GitHub Desktop

Fetch, pull, commit and push.

Keep it open next to Positron.

Tip: under Options → Integrations you can set an external editor; if Positron is not listed, simply open the folder from Positron.

Pull, commit, push

Part 5 of 18 · the everyday cycle, undoing mistakes, what never goes in

The everyday cycle

1 Fetch origin → 2 Pull origin → 3 Work in Positron → 4 Commit to main → 5 Push origin

Fetch origin	asks GitHub what is new; changes nothing on your laptop
Pull origin ↓ 3	downloads 3 new commits into your folder
Commit to main	saves a snapshot of your checked files, on your laptop only
Push origin ↑ 1	uploads your 1 new commit to GitHub

Course repository: only fetch and pull, before every session. Your own work goes into your own folder or repository.

Pull: get the latest version

Before every session

1. **Current repository:** pick the course repository
2. **Click** Fetch origin
3. **If it shows Pull origin ↓:** click it
4. **Done:** the button says Fetch origin again

Pull refused?

- You edited a file that the update changes
- Changes tab: right-click it → Discard changes
- Then click Pull origin again

Golden rule: never edit files inside the course repository. Copy them into your own project folder first.

Commit: save a snapshot

In the Changes tab

1 **Look** at the list of changed files

2 **Click a file** to see the diff: red removed, green added

3 **Tick** the files that belong together

4 **Summary:** what changed, e.g. Add Austria sales chart

5 **Click** Commit to main

Good commits

- Small: one step per commit
- Summary in the imperative: Add, Fix, Update
- Commit when something works
- Copilot button: draft message, then check

History tab: every commit with author, date and diff. You can always go back.

Push: upload your commits

Push

1 **After committing, click** Push origin ↑

2 **New commits on remote?** click Fetch, then Pull origin

3 **Then** Push origin again

4 **Check** on github.com: Repository → View on GitHub

100 MiB
largest file GitHub accepts

50 MiB
GitHub warns from this size

Asked to create a fork? you tried to push to a repository you cannot write to, such as the course repository. Cancel, and keep your work in your own repository.

Oops: undo and fix

Situation	In GitHub Desktop
Changed a file, want the old version	Changes tab: right-click the file → Discard changes
Last commit was wrong, not pushed yet	Click Undo next to the commit message
A pushed commit was wrong	History: right-click the commit → Revert changes in commit
Merge conflict after a pull	Open the file, keep the right lines, delete <<<<<<< ======= >>>>>>>, commit
Folder moved or renamed	Click Locate… and pick the folder



Lost? Do not delete the folder. Ask first: almost everything can be recovered.

What never goes into a repository

Secrets

Passwords, API keys, tokens

Files such as .env

Personal data

Names, e-mails, survey answers

Licensed or confidential data

Big or generated

The .venv folder

Files over 50 MiB, rendered _files

.gitignore (example)

```
.venv/
.env
__pycache__/
*_files/
.DS_Store
```

Ignore a file: Changes tab → right-click → Ignore file

Edit the list: Repository → Repository settings → Ignored files

Pushed a secret? Change the password or key at once

Private or public?

Part 6 of 18 · your own repository, inviting your group, changing visibility

Private and public repositories

	Private	Public
Who can see it	you and people you invite	everyone on the internet
Who can clone it	you and invited people	anyone, without an account
Who can push	you and invited collaborators	you and invited collaborators
Cost on GitHub Free	free, unlimited	free, unlimited
Use it for	group work, assignments, drafts	portfolio, open-source, the course materials

Rule for this course: keep group and assignment repositories private. Never publish data you received under licence or solutions others could copy.

Create your own repository and publish it

1 Create on your laptop

1. **File →** New repository
2. **Name:** e.g. ima-group-3, no spaces
3. **Local path:** Choose… your wu folder
4. **Tick** Initialize with a README
5. **Git ignore:** Python; click Create

2 Publish to GitHub

1. **Click** Publish repository
2. **Check** name and description
3. **Keep this code private:** leave it ticked
4. **Organization:** None (your account)
5. **Click** Publish repository

Existing folder? File → Add local repository, create a repository there, then publish.

Invite your group members

Owner: on github.com

1. **GitHub Desktop:** Repository → View on GitHub
2. **Click** Settings (the repository's, top right)
3. **Left menu:** Collaborators
4. **Click** Add people, type the username
5. **Click** Add … to this repository

Members

- Accept the invitation: e-mail or github.com/notifications
- Clone the repository (Part 4)
- Pull before you start, push when you stop
- One person per file where possible

Invitations expire after seven days; the owner can resend them.

Change visibility later

On github.com

1. **Open** the repository → Settings
2. **General:** scroll to the Danger Zone
3. **Click** Change visibility
4. **Confirm** with the repository name

Before you go public

- The whole history becomes visible, not just today's files
- Anyone can copy and fork it, for good
- Check: no secrets, no personal or licensed data

Public to private: stars and watchers are erased, and copies others already forked stay public.

When GitHub Desktop goes wrong

Message or problem	Fix
Authentication failed / Bad credentials	Options or Settings → Accounts: sign out, sign in again
Push rejected: new commits on remote	Fetch origin, Pull origin, then Push origin
File over 100 MiB	Remove it from the commit; add it to .gitignore
Clone failed in OneDrive or a redirected folder	Clone into a local folder such as C:\Users\you\wu
macOS: username or passphrase not correct	Keychain Access: lock, then unlock 'login'; sign in again
Cannot find the repository	Click Locate… and choose the folder

Log files: Help → Show Logs in Explorer (Windows) or Show Logs in Finder (macOS); send them when you ask for help.

Checklist: GitHub

1. **Account** with 2FA and recovery codes stored
2. **E-mail:** WU address verified, Keep my email addresses private on
3. **GitHub Education** application submitted
4. **GitHub Desktop** installed, signed in, noreply address under Git
5. **Course repository** cloned to a local folder and opened in Positron
6. **Group repository** created, private, members invited (optional)

Get GitHub Copilot

Part 7 of 18 · Posit Assistant, what Copilot does, the free plans, privacy settings

Positron Assistant is now Posit Assistant

2025.07
Positron Assistant (preview)

2025.10
Copilot also for chat

2026.04
Posit Assistant (preview)

2026.07
Posit Assistant default; old one removed

2026.10
current Positron release

Positron: the free editor by Posit, where you write and run Python

Posit Assistant: the AI chat built into Positron; it has no model of its own

GitHub Copilot: the model provider: it answers in the chat and suggests code

Old tutorials? Ask/Edit/Agent modes or positron.assistant.enable mean the guide is outdated.

What Copilot does in Positron

Provider	Chat	Completions	Cost for you
GitHub Copilot (Preview)	yes	yes, ghost text	free with Copilot Student or Free
Posit AI Pass	yes	yes	20 USD a month
Anthropic, OpenAI, others	yes	no	paid API key

Chat

Ask, plan, run code after your OK, explain errors

Completions

Grey suggestions while you type; Tab accepts

For this course: GitHub Copilot. It costs nothing with GitHub Education.

Two free Copilot plans

	Copilot Free	Copilot Student
Who	anyone with a GitHub account	verified students (GitHub Education)
Code completions	2,000 a month	unlimited
Chat and agent	an allowance of AI credits	a larger allowance of AI credits
Models	automatic selection only	automatic selection only
Price	free	free while you study

Credits run out? they reset on the 1st of the month. The Copilot icon in Positron's status bar shows how much you have used.

Activate Copilot Student

Copilot Student

1. **Be verified** in GitHub Education (Part 2)
2. Open github.com/settings/education/benefits
3. **Under Free GitHub developer resources:** Learn more
4. **Follow the prompts** to activate Copilot Student
5. **Check** github.com/settings/copilot

Not approved yet?

- Use Copilot Free meanwhile
- github.com/settings/copilot → Start using Copilot Free
- Shown a paid checkout? Do not pay; wait a few days

Re-checked monthly: the free plan ends when you are no longer a student.

Privacy settings on github.com

Opt out of model training

1 Open github.com/settings/copilot

2 **Find** Allow GitHub to use my data for AI model training

3 **Set it to** Disabled

4 **Done:** the setting saves at once

Posit does not see your prompts

GitHub / Microsoft receive prompt and context

Their terms apply to what you send

Rule: send code and anonymised data only. Names, e-mails, customer IDs and confidential files never go into a prompt.

Connect Copilot to Positron

Part 8 of 18 · check the version, sign in, test that it works

Before you start

1. **Positron 2026.07 or later:** Help → Check for Updates (current: 2026.10)
2. **Open your project folder:** File → Open Folder
3. **Trust the folder** when Positron asks: Yes, I trust the authors
4. **Python running:** pick Python 3.14 (.venv) at the top right
5. **GitHub account** with Copilot Free or Copilot Student active

Nothing to switch on: AI is ready; it waits for a provider.

Connect GitHub Copilot

In Positron

1. **Command Palette:** Ctrl/Cmd + Shift + P
2. **Type and run:** Authentication: Configure Language Model Providers
3. **Pick** GitHub Copilot
4. **Sign in** in the browser, click Authorize
5. **Back in Positron:** Copilot shows as connected

Command Palette

```
Authentication:
  Configure Language
  Model Providers
```

Shortcut

- Accounts menu (person icon, bottom left)
- → Configure Language Model Providers

Signed in to GitHub, but no Copilot? you must use Configure Language Model Providers.

Check that it works

Connected: the provider dialog shows GitHub Copilot as connected

Chat: View: Show Posit Assistant opens; Copilot models in the model picker

Ghost text: typing in a .py file or a Quarto code cell shows grey suggestions

Status bar: the Copilot icon at the bottom shows your usage

Report: Command Palette → AI: Create Diagnostic Report

All five fine? you are ready.

If the browser does not return to Positron

Sign in with a device code

1. **Settings:** search preferDeviceCodeFlow, tick it
2. **Run again:** Configure Language Model Providers
3. **Dialog shows** Your Code: e.g. 4F2A-91C7
4. **Click** Copy & Continue to Browser
5. **Paste the code** on GitHub, click Authorize

Still stuck?

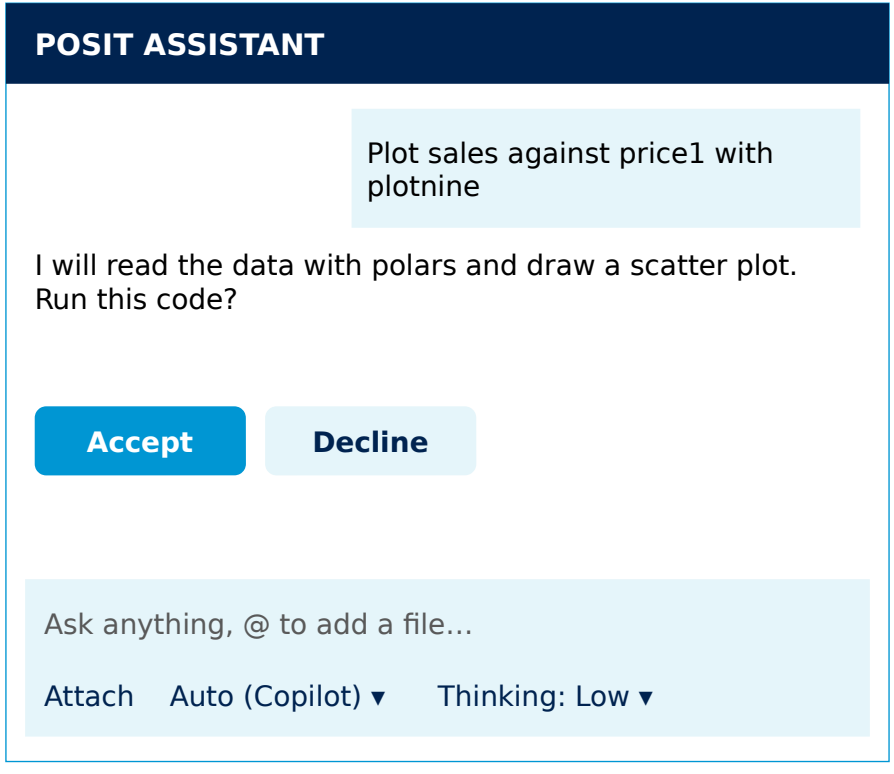
- Accounts → Manage Accounts: sign out, sign in again
- Check Copilot is active on github.com/settings/copilot
- Restart Positron

University or company network? proxies can block the redirect; the device code usually works.

Chat with Posit Assistant

Part 9 of 18 · the pane, what it sees,
permissions, slash commands, Fix and Explain

The Posit Assistant pane



1 Open it
activity-bar icon or View: Show Posit Assistant

2 Your request
plain English; name files, packages, outputs

3 Accept / Decline
nothing runs without your OK

4 @ and paperclip
add files, images or the .qmd you work on

5 Model and thinking
Auto with Copilot; more thinking = slower, more careful

What Posit Assistant can see

Your session

- Python version
- Variable names and types
- Data-frame columns and shape

Your output

- Plots
- Console history
- Errors

Your files

- Open files and selection
- Project folder structure
- Files you attach

Tip: load the data first, then ask; the assistant sees the column names and gets them right

Keep files out: setting positron.assistant.aiExcludes, e.g. data/personal/**

It sees what you see: never load personal data you may not share.

It asks before it runs code

Mode	What happens	For you
Normal (default)	safe actions run; risky ones ask first	use this
Restricted	every action asks first	fine, automatic in untrusted folders
Plan (/plan)	writes a plan, changes nothing until you approve	good for bigger tasks
Ask (/ask)	answers only, read-only	good for questions
Auto, YOLO	approves actions automatically	do not use

Before you click Accept: read the code. Does it use polars, the right file and the right columns? If not, Decline and say what to change.

Slash commands, Fix and Explain

Type	What it does
/new	start a new conversation (new topic)
/plan	plan first, code later
/ask	questions only, no changes
/report	turn the chat into a Quarto .qmd
/notebook	export the chat to a notebook

Error in the Console

```
ColumnNotFoundError:
  unable to find column "Sales"
```

Fix **Explain**

Explain first: understand the error, then fix it yourself or with Fix.

Learning tip: use Explain more than Fix. You will meet the same error in the exam, without the assistant.

Example: the chocolate data

Your prompt

```
Use polars, not pandas, to read
data/legacy/chocolate_dataset.csv.
Plot sales against price1 with
plotnine. Fit sales ~ price1 with the
statsmodels formula API. Explain
each step in two sentences.
```

Then

1. **Read** the plan and code
2. **Accept** if it fits
3. **Check:** slope ≈ −669, R^2 ≈ 0.88
4. **Ask** why, if unsure

Names the file so it reads the right data

Names the packages so the code matches the course: polars, plotnine, statsmodels

✓ **Same numbers as on the slides?** then the code is right.

Code completions

Part 10 of 18 · ghost text, keys, switching it on and off

Ghost text: accept, take part, or ignore

```
choc = pl.read_csv(path)
model = smf.ols("sales ~ price1", data=choc
.to_pandas()).fit()
```

Grey text = Copilot's suggestion, not yet in your file

Key	Effect
Tab	accept the whole suggestion
Ctrl/Cmd + →	accept word by word
Esc or keep typing	ignore it
**Alt/Option + **	ask for a suggestion

Tab is not a reflex: read the grey text first. It guesses from patterns and can pick the wrong column, file or method.

Where: .py files and Quarto code cells; not in plain text or Markdown prose

Switch completions on and off

Quick: the status bar

1. **Click** the Copilot icon at the bottom
2. **Choose:** off for this file type, or all files
3. **Or:** Snooze Inline Suggestions for a while

settings.json: off for Python only

```
"github.copilot.enable": {
  "*": true,
  "python": false,
  "markdown": false
}
```

Learning a new method? turn completions off while you practise; switch them on when you work on the case study.

Use it well

Part 11 of 18 · good prompts, checking results, data protection and course rules

Prompts that work

Weak

- “Analyse my data.”
- “Make a chart.”

Strong

- “Use polars to read data/case/orders.csv and plot weekly orders by country with plotnine.”

1. **Name the file, the packages and the output** you want
2. **Say polars** explicitly: the assistant often defaults to pandas
3. **Ask it to explain** each step, or the assumptions of a model
4. **One topic per chat:** /new for the next one

Check everything

Run it

- Run the code yourself
- Does it render from top to bottom?

Compare

- Numbers against the output
- Columns against the data

Understand

- Can you explain each line?
- Is it the method we learned?

Typical slips: pandas instead of polars, a wrong column, the wrong test, invented numbers

Numbers in text: always from inline code, never typed from the chat

You own every number: the assistant cannot be blamed in a report or an exam.

Data protection and course rules

Never in a prompt

- Names, e-mails, student or customer IDs
- Passwords, API keys
- Confidential or licensed data
- Other students' work

Course rules

- AI use is allowed and encouraged
- Prompt log: exercises and case study appendix
- You verify and own every result
- Exam: closed book, no AI

Prompt log: copy the prompts that produced non-trivial code, and note what you kept or changed.

Fix problems

Part 12 of 18 · common problems, outdated instructions, checklist

When Copilot does not work

Problem	Fix
No Posit Assistant	Update Positron (2026.07+); trust the folder; View: Show Posit Assistant
Signed in to GitHub, no Copilot	Configure Language Model Providers → GitHub Copilot
No ghost text	status-bar icon: not snoozed or off; file is .py or a code cell
Usage limit reached	wait until the 1st; activate Copilot Student
Paid checkout on github.com	do not pay; wait a few days, try github.com/settings/copilot
Odd answers	add context (files, packages), or start again with /new

Diagnose: Command Palette → AI: Create Diagnostic Report.

Old instructions, and what to use now

If a guide says	Use now
Positron Assistant	Posit Assistant
Setting positron.assistant.enable	nothing to set (assistant.enabled is on)
Ask / Edit / Agent modes	Normal, /ask, /plan
Positron Assistant: Collect Diagnostics	AI: Create Diagnostic Report
Inline chat with Ctrl/Cmd + I	the Posit Assistant pane

Check the date: AI tools change monthly. Prefer positron.posit.co and GitHub Docs over older blog posts.

Checklist: Copilot

1. **Copilot Free or Student** active on github.com/settings/copilot
2. **Model training** opt-out set to Disabled
3. **Positron 2026.07 or later,** project folder trusted
4. **Copilot connected:** Configure Language Model Providers
5. **Posit Assistant** answers, and ghost text appears in a code cell
6. **Prompt log** started for the coding exercise

Claude and ChatGPT: the four routes

Part 13 of 18 · optional: subscription or API key,
which route gives what

Four routes at a glance

Route	What you get	Claude needs	ChatGPT needs
1 Posit Assistant	chat in Positron's own AI pane	API key	API key
2 Vendor extension	an agent panel in Positron	Pro or Max	ChatGPT account
3 Terminal	the agent in Positron's terminal	Pro or Max	ChatGPT account
4 MCP (experimental)	agent works in your live session	with route 2 or 3	with route 2 or 3



No code completions: the grey suggestions while typing come only from GitHub Copilot or Posit AI Pass, not from Claude or ChatGPT.

Subscription or API key: the key question

Subscription (monthly)

- Claude Pro or Max, ChatGPT Plus or Pro
- Works in: extension and terminal (routes 2, 3)
- Not in Posit Assistant
- Flat price, usage limits

API key (pay per use)

- From platform.claude.com or platform.openai.com
- Works in: Posit Assistant, also routes 2 and 3
- Prepaid credit, billed per token
- Separate from any subscription

Common surprise: “I pay for Claude Pro, why does Positron want a key?” Because Pro does not include API access.

Route 1: Posit Assistant with an API key

Part 14 of 18 · get a key, connect it, what it costs

Get an Anthropic (Claude) API key

Claude Console

1. Open platform.claude.com
2. **Settings → Billing:** buy a small credit, e.g. 5 USD
3. **Set a spend limit;** auto-reload off
4. **Settings → API keys:** Create Key, then Add
5. **Copy the key** into a password manager

Good to know

- Credits expire after one year
- Claude Pro does not include API credit
- API use is not used for training

The key is a password: never in code, a .qmd file, GitHub or a chat.

Get an OpenAI (ChatGPT) API key

OpenAI platform

1. Open platform.openai.com
2. **Billing:** add prepaid credit, at least 5 USD
3. **Switch off** auto-recharge; set a budget
4. **API keys:** Create new secret key
5. **Copy the key** into a password manager

Good to know

- ChatGPT Plus does not include API credit
- Any OpenAI-compatible service also works
- API use is not used for training by default

Leaked a key? delete it on the platform at once and create a new one.

Connect the key in Positron

In Positron

1 **Command Palette:** Ctrl/Cmd + Shift + P

2 **Run:** Authentication: Configure Language Model Providers

3 **Pick** Anthropic or OpenAI

4 **Paste** the API key; it shows as connected

5 **View: Show Posit Assistant:** pick a model

Command Palette

```
Authentication:
  Configure Language
  Model Providers
```

Stored safely: in the system keychain, not in a file

Models: whatever your key may use

Same pane as Copilot: Parts 9 and 11 apply unchanged.

What a question costs

Claude model	Input (per 1M tokens)	Output (per 1M tokens)	One question*
Haiku 5.5	0.10 USD	0.50 USD	≈ 0.1 cent
Sonnet 5.5	2 USD	10 USD	≈ 2 cents
Opus 5.5	4 USD	20 USD	≈ 4 cents

* 5,000 tokens in, 1,000 out; own calculation from Anthropic's prices.

Chat is cheap, agents are not: an agent that reads your whole project can cost dollars

Protect yourself: small prepaid credit, a spend limit, no auto-reload

Free alternative: GitHub Copilot Student in the same pane.

Route 2: the vendor's extension

Part 15 of 18 · Claude Code and Codex as panels in Positron

Claude Code extension

Install and sign in

1 **Extensions:** Ctrl/Cmd + Shift + X, search Claude Code

2 **Check:** publisher Anthropic, verified badge; Install

3 **Open** your project folder and trust it

4 **Open a file, click** the spark icon or ✻ Claude Code

5 **Sign in** with Claude Pro or Max in the browser

Good to know

- Free claude.ai plan: no Claude Code
- No spark icon? Open a file, Developer: Reload Window
- Shows changes as diffs you accept

Not officially supported in Positron: if it fails, use the terminal (Part 16).

Codex extension (OpenAI)

Install and sign in

1. **Extensions:** search Codex
2. **Check:** publisher OpenAI, ID openai.chatgpt
3. **Click** Install and trust the publisher
4. **Open** the Codex panel
5. **Sign in** with your ChatGPT account

Good to know

- Works with ChatGPT Free, Go, Plus and higher; limits differ
- Same agent as the Codex CLI
- Windows: native, or WSL if needed

Not in the list? use the Codex CLI in the terminal (Part 16).

If an extension does not show up

1. **Search again** with the publisher name: Anthropic or OpenAI
2. **Switch the gallery:** setting positron.extensions.gallerySource → Open VSX, restart
3. **Install a file:** Extensions: Install from VSIX
4. **Or skip it:** the terminal route needs no extension

Only official publishers: fake look-alike extensions exist. Check the publisher and the verified badge.

Route 3: the agent in the terminal

Part 16 of 18 · install Claude Code and Codex, sign in, work with an agent

Install Claude Code in the terminal

Windows: Positron terminal (PowerShell)

```
irm https://claude.ai/install.ps1 | iex
```

macOS: Positron terminal

```
curl -fsSL https://claude.ai/install.sh | bash
```

New terminal: check, then start

```
claude --version
claude
```

Sign in

- Pro, Max or Console account
- Code shown? Paste it

Billed to the API by mistake? if ANTHROPIC_API_KEY is set, Claude Code may use it instead of Pro. Type /status to check.

Install Codex in the terminal

Windows: Positron terminal (PowerShell)

```
powershell -ExecutionPolicy ByPass -c "irm https://chatgpt.com/codex/install.ps1 | iex"
```

macOS: Positron terminal

```
curl -fsSL https://chatgpt.com/codex/install.sh | sh
```

New terminal: start, check

```
codex
codex login status
```

Sign in

- Choose Sign in with ChatGPT
- Browser fails? codex login --device-auth

Works without the extension: the terminal route is the most reliable way in Positron.

Working with a coding agent

Open the project → **Start the agent** → **Ask in plain English** → **Approve changes** → **Run and check**

CLAUDE.md or AGENTS.md

```
Use polars, plotnine, great_tables
and statsmodels only. Explain each
model in two sentences. Never
read data/personal/.
```

A good request

```
Read report.qmd and list every
number in the text that is not
computed by inline code.
```

Ask-first mode: read every proposed change and command before you approve it

Small steps: one task at a time, then render and check



You own every number: agents make confident mistakes.

Route 4: MCP, the agent in your live session

Part 17 of 18 · what it does, how to switch it on, why to be careful

What Positron's MCP server does

Run code

- in your live Python session
- labelled in the Console

See plots

- the current plot
- as an image

Use Positron

- run Positron commands
- read session history

Agents: Claude Code, Codex and Gemini CLI

Why useful: the agent sees your real data frame instead of guessing its columns

Experimental: off by default; settings and behaviour may change.

Switch it on

Claude Code

1. **Settings:** tick ai.mcp.enabled
2. **Open** the project, start Python
3. **Terminal:** claude (restart if open)
4. **Type** /mcp: positron is listed

Codex

1. **Settings:** tick ai.mcp.enabled
2. **Command Palette:** Positron MCP: Add to Coding Agent → Codex
3. **Positron terminal:** codex
4. **Check:** Positron MCP: Show Connected Agents

Then ask: “Use the positron tools to summarise the data frame choc in my Python session.”

Why to be careful with it

1. **No approval in Positron:** only the agent's own prompts stand between it and your session
2. **It sees what you loaded:** never load personal data in a session an agent can use
3. **Keep ask-first mode:** never allow every command automatically
4. **Switch it off** when you do not need it: untick ai.mcp.enabled

⚠ **For this course:** a demonstration, not a requirement. Coursework never depends on it.

Choose and use it safely

Part 18 of 18 · which route for what, costs, privacy, course rules, problems

Which route for what

I want …	Best route	Cost for a student
chat about my data, free	Posit Assistant + GitHub Copilot Student	0 USD
chat with a specific Claude or GPT model	Posit Assistant + API key	pay per use
an agent for bigger tasks, I have Claude Pro	Claude Code: terminal or extension	20 USD a month (Pro)
an agent, I have a ChatGPT account	Codex: terminal or extension	0–20 USD a month
grey code suggestions while typing	GitHub Copilot or Posit AI Pass	0 or 20 USD a month

Course default: Posit Assistant with GitHub Copilot Student. Everything else is optional.

Privacy: who sees your prompts

API key (routes 1 to 3)

- Not used for training by default
- Kept about 30 days
- Posit sees nothing

Subscription (Pro, Plus, Free)

- May be used for training while the setting is on
- Claude: claude.ai/settings/data-privacy-controls
- ChatGPT: Settings → Data Controls

Switch training off in your Claude and ChatGPT settings before you use them for coursework.

API keys, data and course rules

Never in a prompt or a file

- API keys and passwords
- Names, e-mails, student or customer IDs
- Confidential or licensed data
- Exam or assessment content

Course rules

- Course default: GitHub Copilot
- Nobody has to pay for anything
- Same rules as in Part 11
- Name the tool and model in the prompt log

Prompt log: note which tool and model you used, the prompt, and what you kept or changed.

When Claude or ChatGPT does not work

Problem	Fix
Claude Pro, but Positron wants a key	expected: use a Console API key, or Claude Code (routes 2, 3)
Key rejected	copy again without spaces; is there credit on the account?
No code completions	not offered by Claude or OpenAI; use GitHub Copilot
claude or codex: command not found	open a new terminal; run the installer again
Claude Code bills the API, not Pro	unset ANTHROPIC_API_KEY; check with /status
Agent cannot see the session (MCP)	ai.mcp.enabled on, Python running, restart the agent

Diagnose: Command Palette → AI: Create Diagnostic Report.

Checklist: Claude and ChatGPT

1. **Route chosen:** Posit Assistant, extension, terminal, or MCP
2. **Account ready:** API key with a spend limit, or Claude Pro / ChatGPT account
3. **Training off** in your Claude or ChatGPT privacy settings
4. **Connected and tested** with one small question about your data
5. **Standing instructions** in CLAUDE.md or AGENTS.md
6. **Prompt log** started

Official guides: GitHub

- GitHub Desktop: getting started
- Cloning a repository in GitHub Desktop
- Committing, pushing and syncing
- Two-factor authentication
- Apply to GitHub Education as a student
- Setting repository visibility
- GitHub Skills: free interactive courses

Official guides: Copilot in Positron

- Posit Assistant in Positron
- Getting started and model providers
- Code completions
- AI configuration and privacy settings
- Set up Copilot for students (GitHub Docs)
- Copilot plans
- Positron release notes

Official guides: Claude and ChatGPT

- Claude Code in VS Code and other editors
- Claude Code: install and set up
- Claude Code: data usage and privacy
- Claude plans and API prices
- Posit Assistant model providers
- OpenAI Codex (repository and install)
- Positron release notes (MCP server)

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